Detección de Anomalías Acústicas en Maquinaria Industrial

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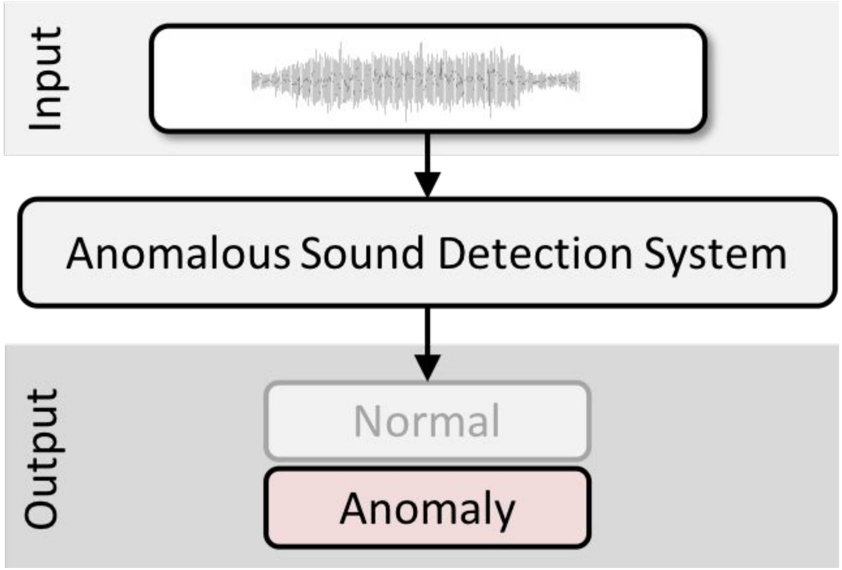
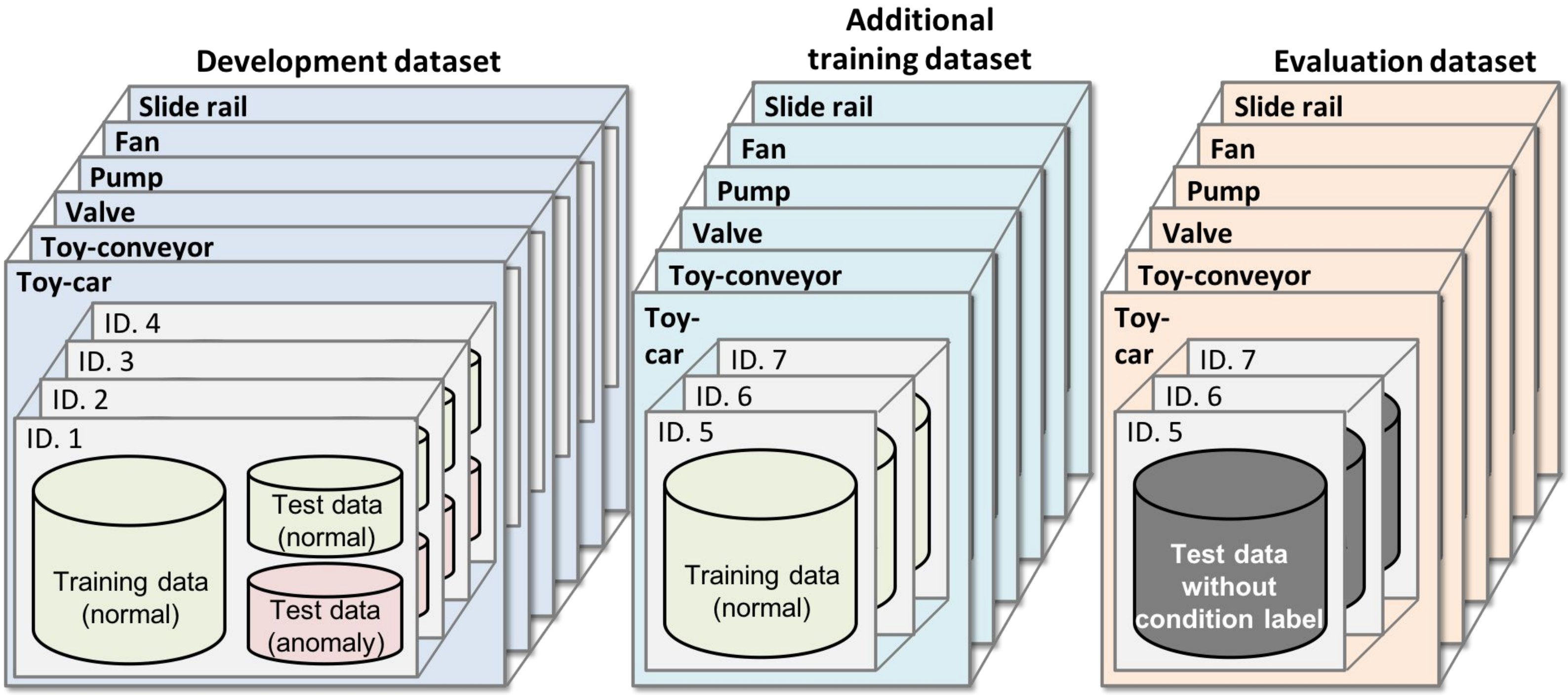
Alcance

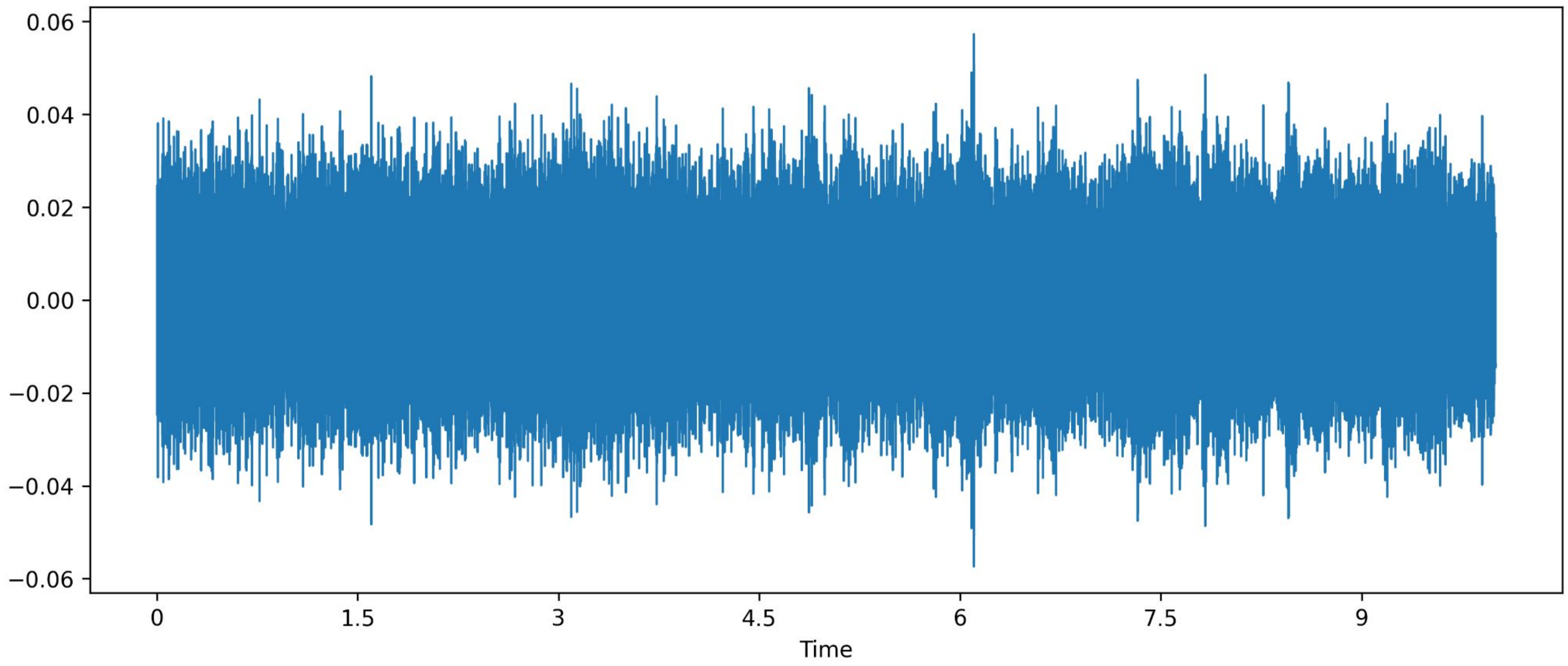


Detectar anomalías acústicas en maquinaria industrial haciendo uso de autoencoders

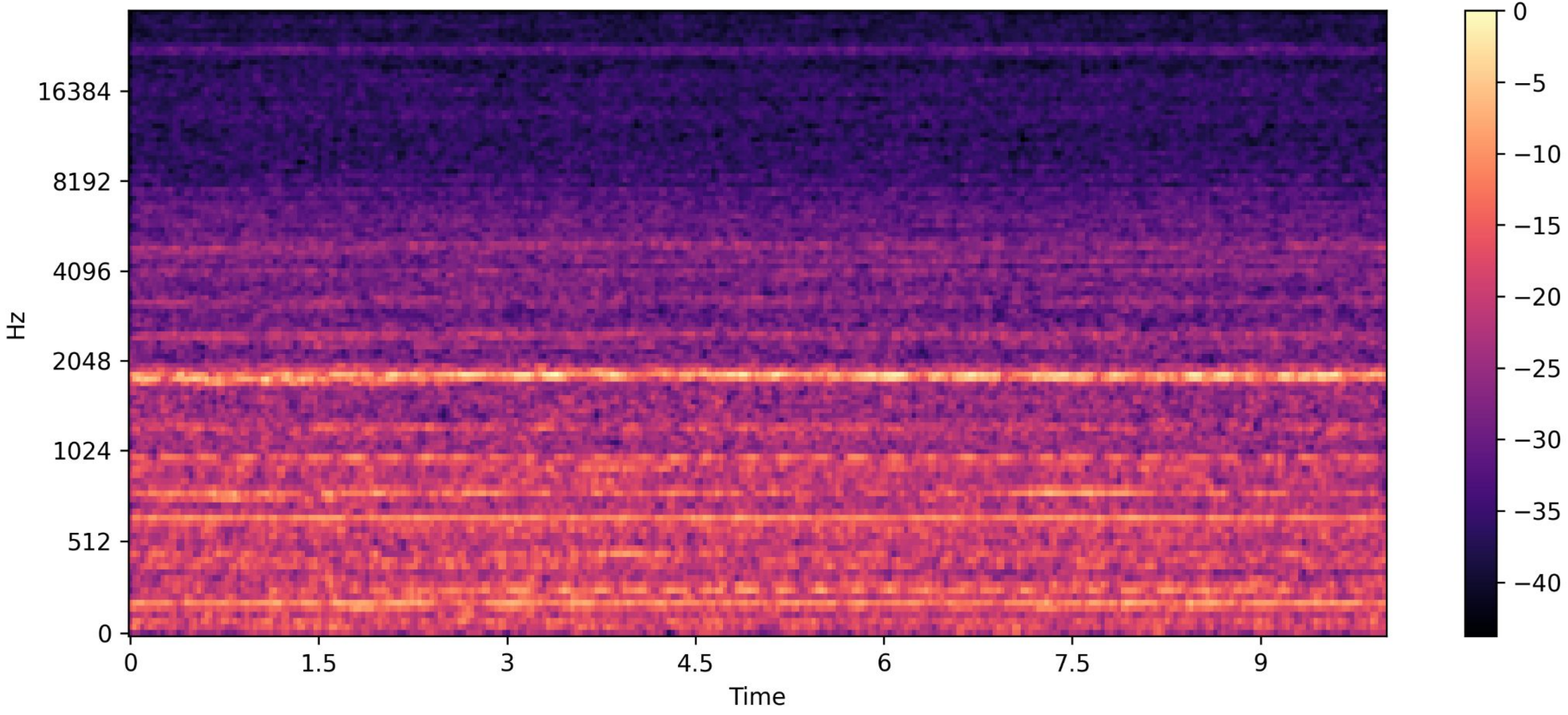
Dataset



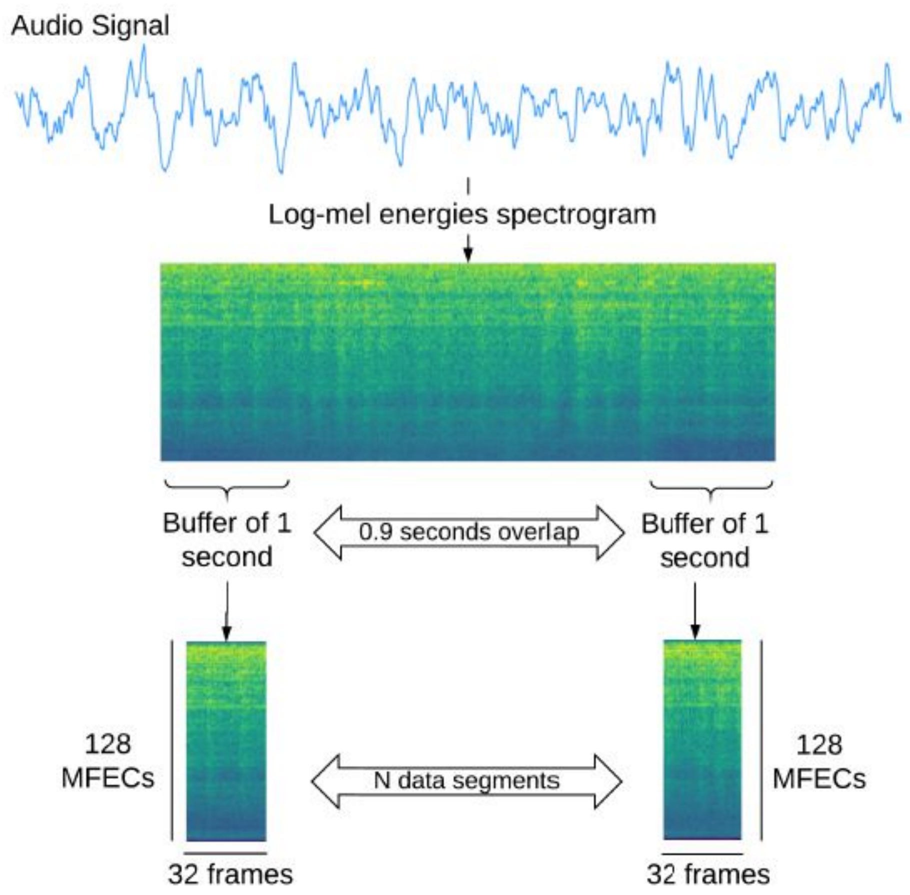
Development dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 4
ID. 3
ID. 2
ID. 1
Training data (normal)
Test data (normal)
Test data (anomaly)
Additional training dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 7
ID. 6
ID. 5
Training data (normal)
Evaluation dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 7
ID. 6
ID. 5
Test data without condition label



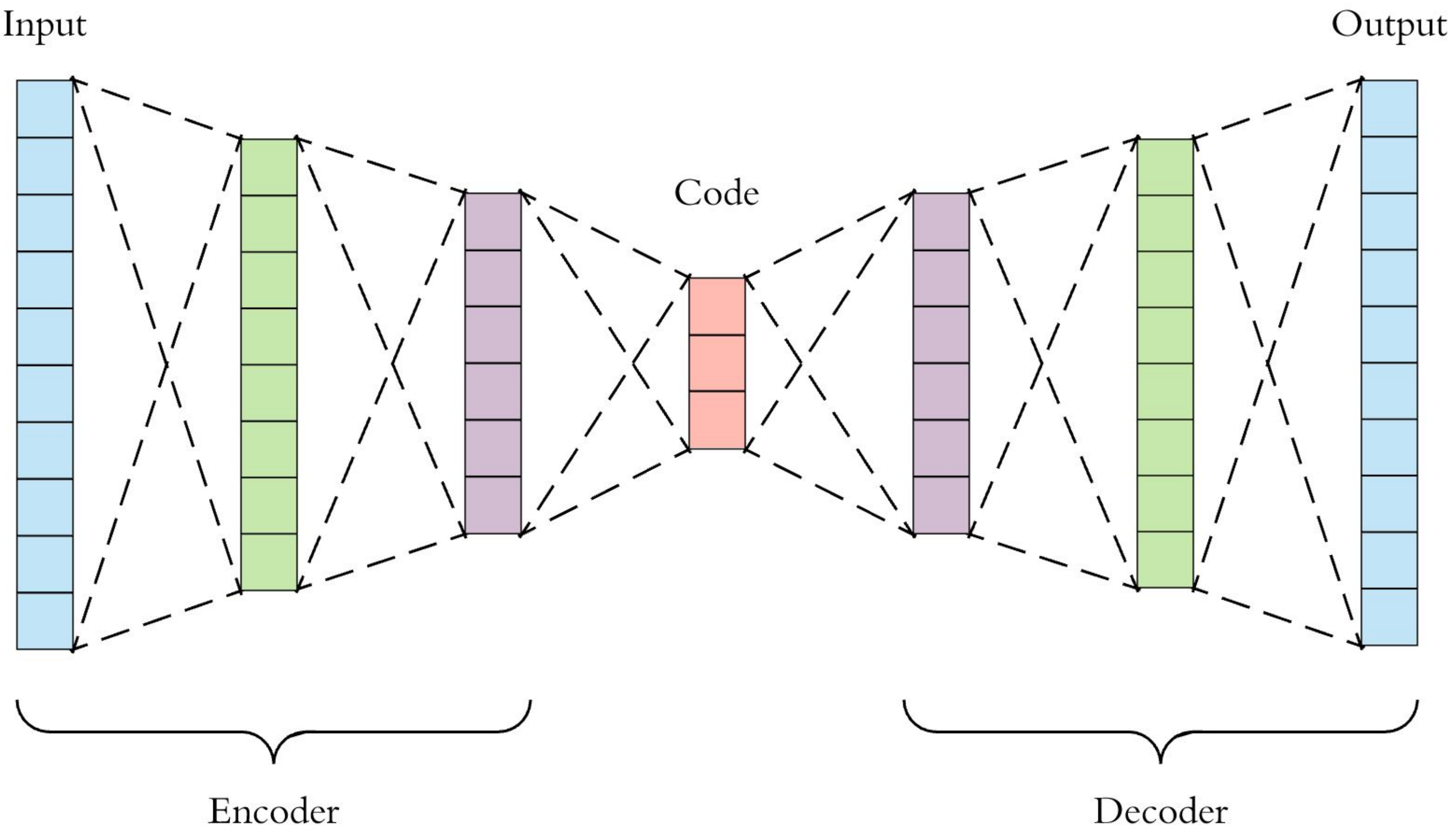
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0
1.5
3
4.5
6
7.5
9
Time



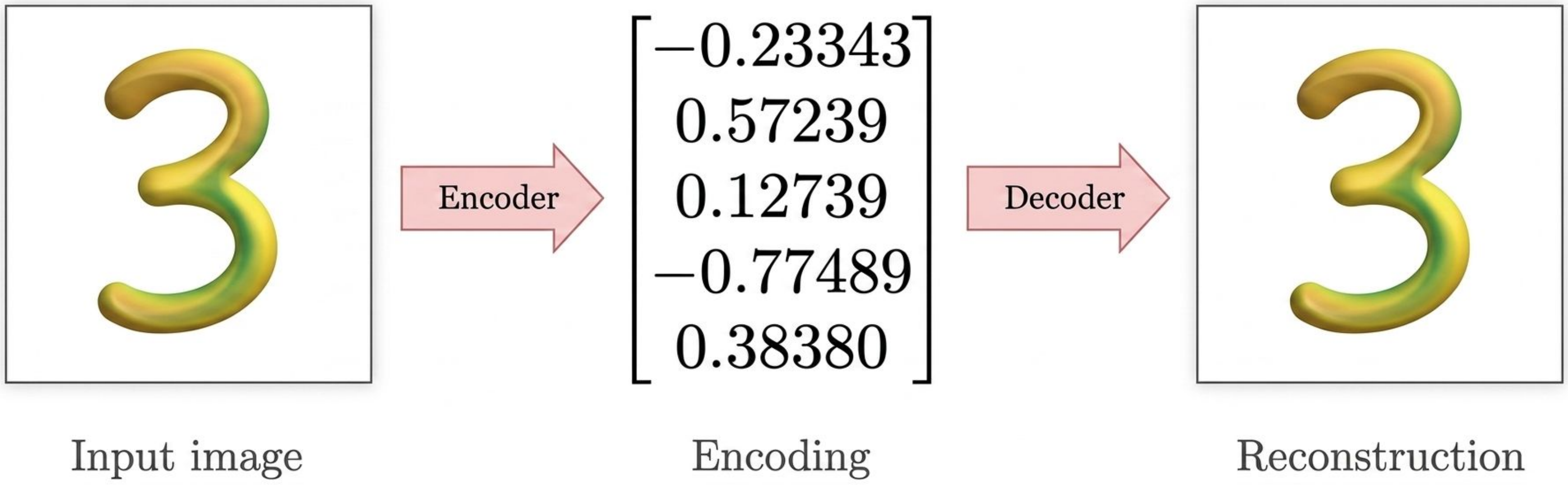
16384
8192
4096
Hz
2048
1024
512
0
0
1.5
3
4.5
6
7.5
9
Time
0
−5
−10
−15
−20
−25
−30
−35
−40



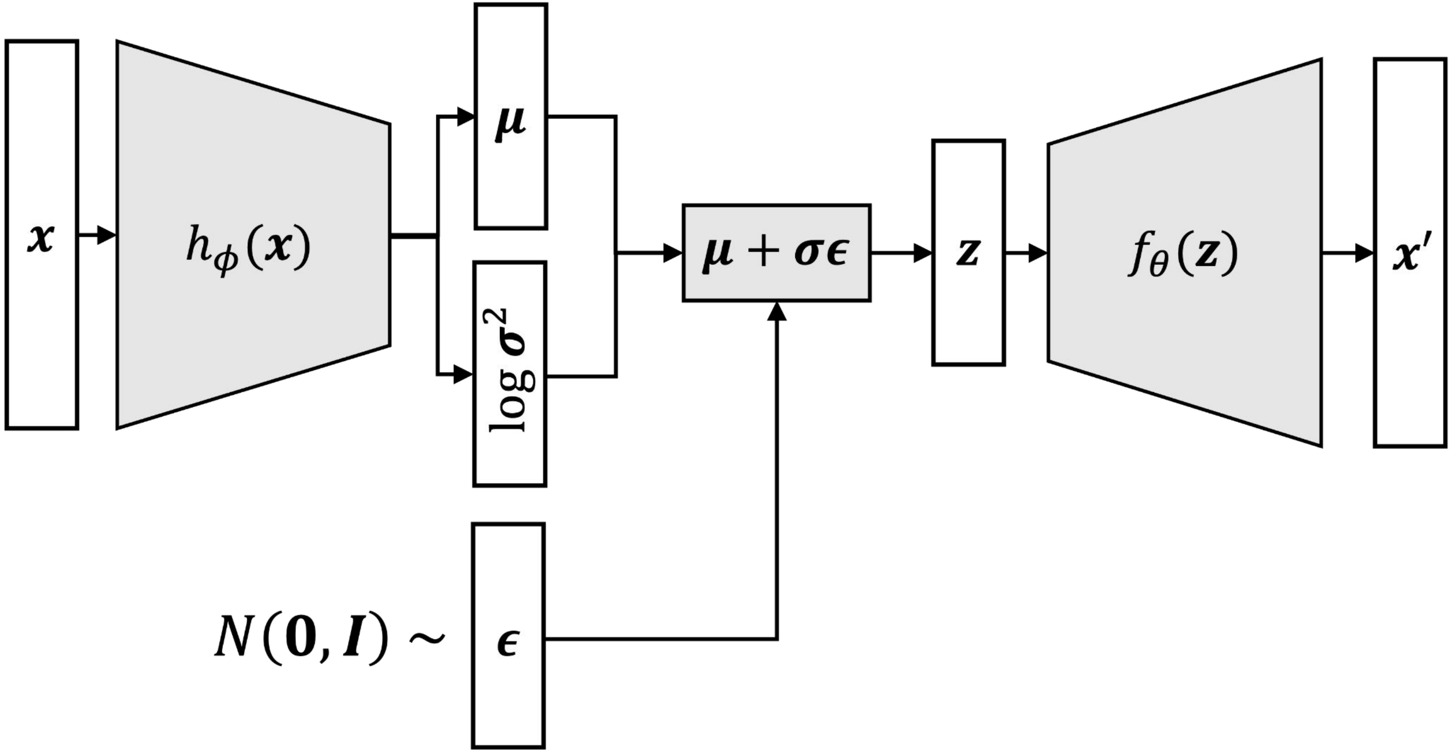
Audio Signal
Log-mel energies spectrogram
Buffer of 1 second
0.9 seconds overlap
Buffer of 1 second
128 MFECs
N data segments
128 MFECs
32 frames
32 frames



Input
Output
Code
Encoder
Decoder


Encoder
−0.23343
0.57239
0.12739
−0.77489
0.38380
Decoder
Input image
Encoding
Reconstruction

Variational Autoencoder



Input image

$$\xrightarrow{\text{Encoder}}$$

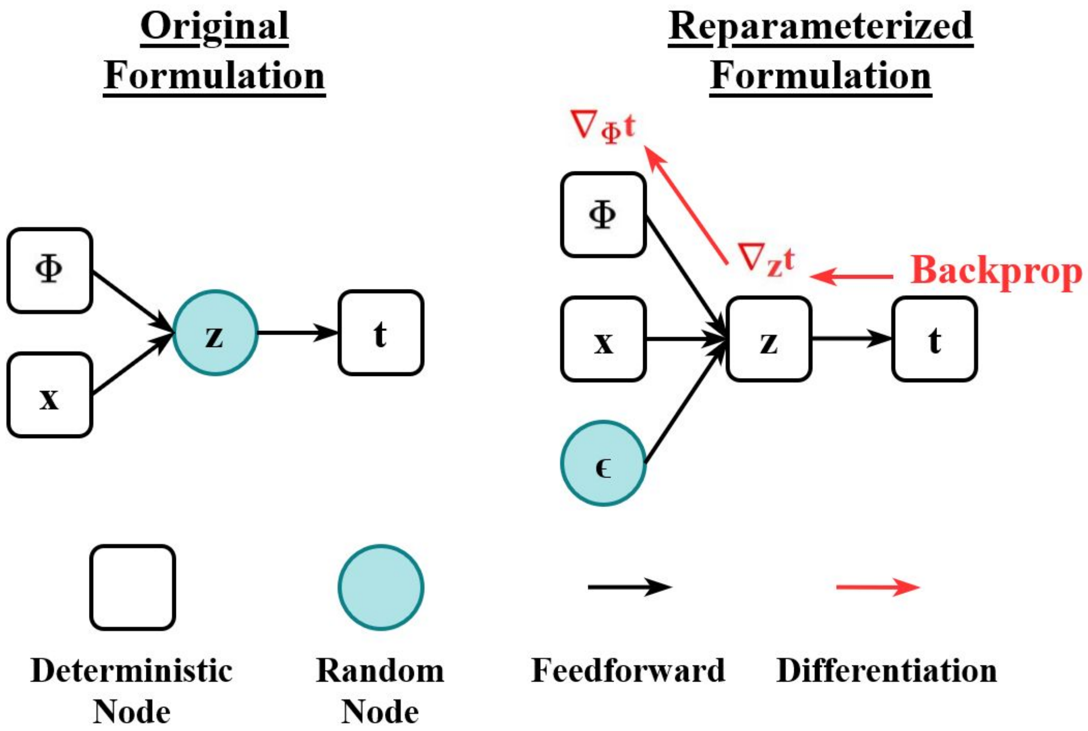
Encoding in 'latent space'

$$\begin{bmatrix} 0.349 \\ 0.929 \\ -0.299 \\ 1.690 \\ 0.380 \end{bmatrix} \quad \begin{bmatrix} 4.12 \\ 2.33 \\ 1.12 \\ 3.24 \\ 2.05 \end{bmatrix}$$

Means
μ

Stddev
σ

$$\begin{bmatrix} X(0.349, 4.12^2) \\ X(0.929, 2.33^2) \\ X(-0.299, 1.12^2) \\ X(1.690, 3.24^2) \\ X(0.380, 2.05^2) \end{bmatrix} \xrightarrow{\text{Sample}} \begin{bmatrix} 0.602 \\ 2.331 \\ -0.001 \\ 6.349 \\ 2.187 \end{bmatrix}$$


Original
Formulation
Reparameterized
Formulation
Φ
x
z
t
∇Φt
∇zt
Backprop
Φ
x
z
t
ϵ
Deterministic
Node
Random
Node
Feedforward
Differentiation

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

$$\epsilon \sim \mathcal{N}(0, I)$$

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

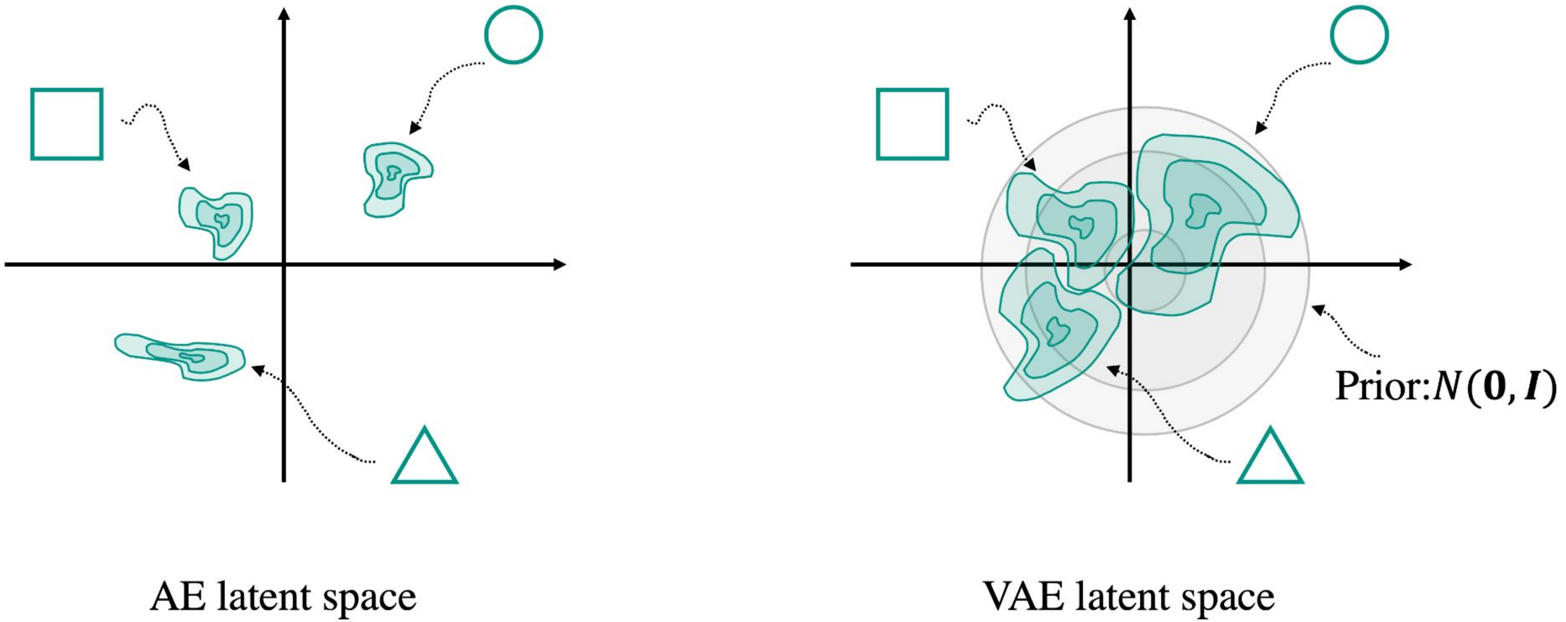
$$\epsilon \sim \mathcal{N}(0, I)$$

$$\sigma \cdot \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

$$\downarrow$$

$$\mu + \sigma \cdot \epsilon \sim \mathcal{N}(\mu, \sigma^2)$$


Prior: $N(\mathbf{0}, \boldsymbol{I})$
AE latent space
VAE latent space

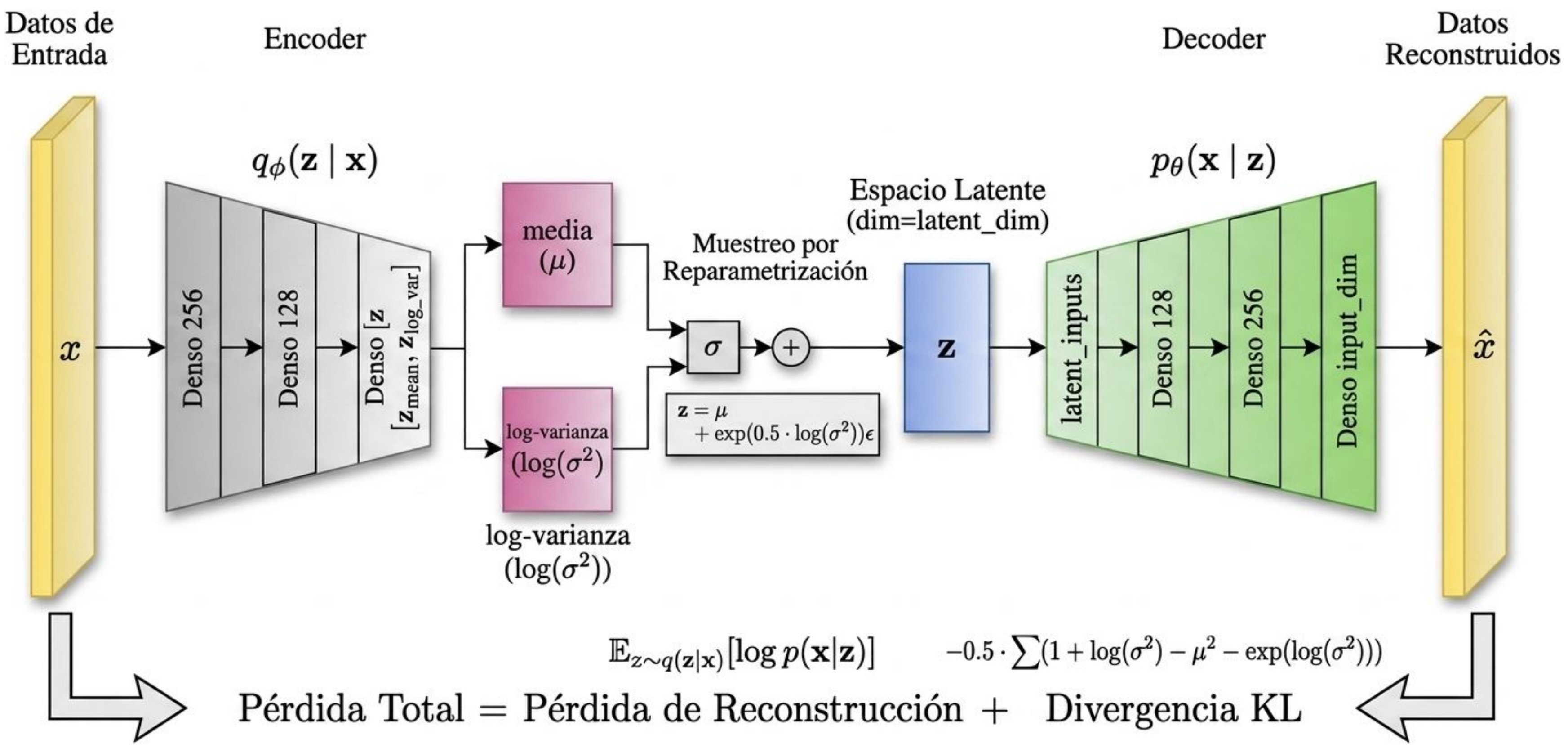
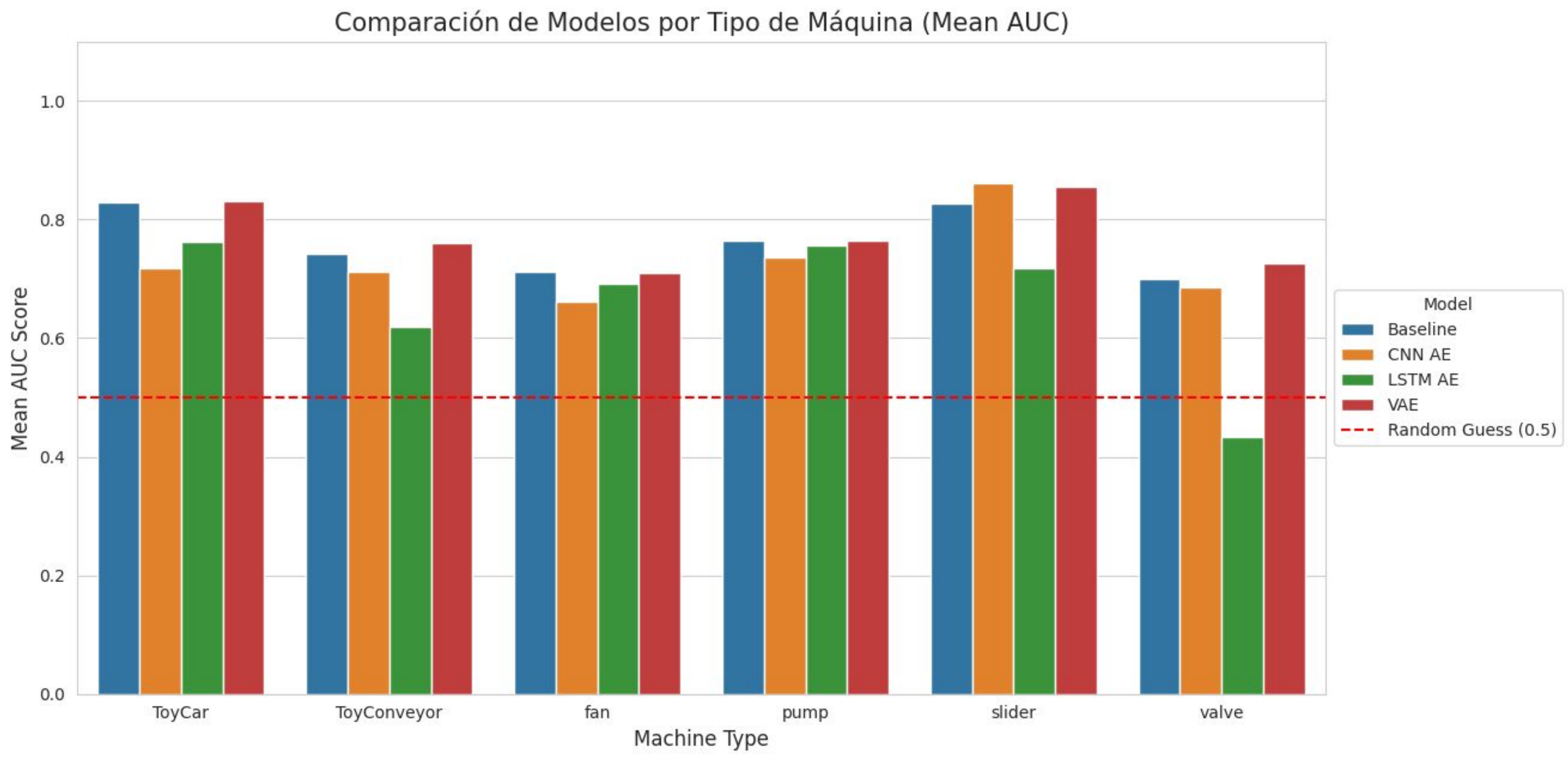
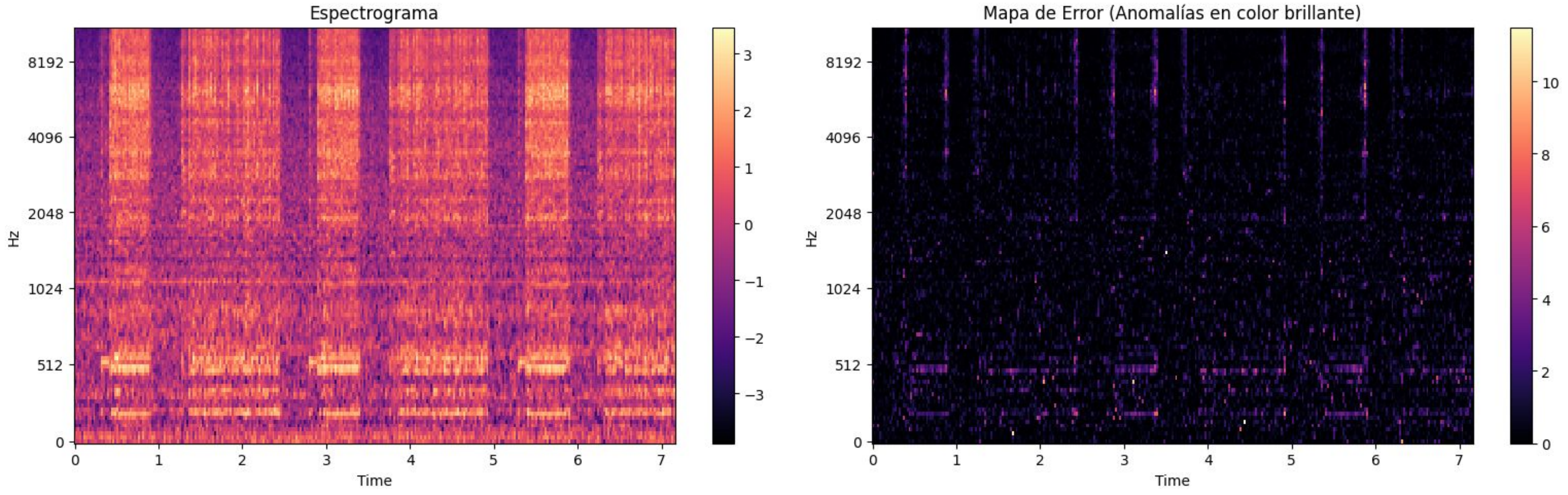


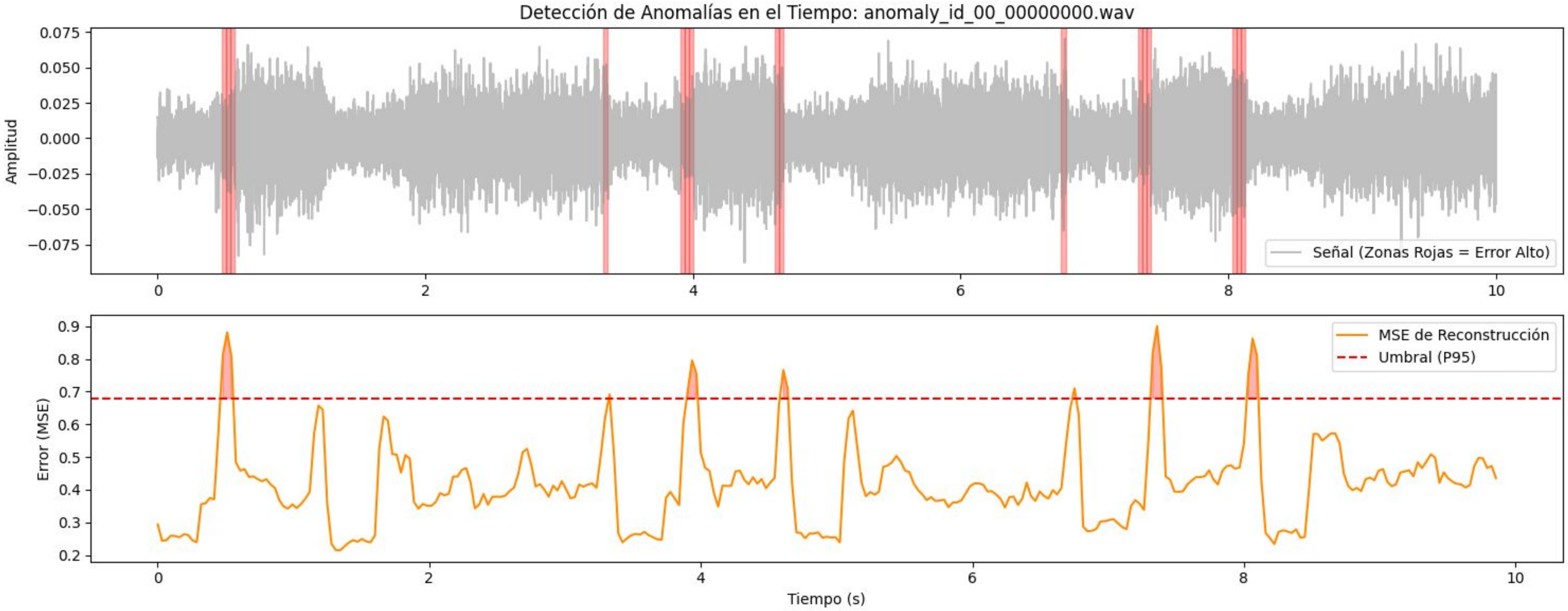
Diagrama de Arquitectura Keras-VAE


Comparación de Modelos por Tipo de Máquina (Mean AUC)
Mean AUC Score
1.0
0.8
0.6
0.4
0.2
0.0
ToyCar
ToyConveyor
fan
pump
slider
valve
Machine Type
Model
Baseline
CNN AE
LSTM AE
VAE
Random Guess (0.5)

Modelo	**Promedio AUC**
AE	0.763728
CNN AE	0.730149
LSTM AE	0.666109
VAE	**0.775723**

Mapa de Errores





Detección de Anomalías en el Tiempo: anomaly_id_00_00000000.wav
Amplitud
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
Señal (Zonas Rojas = Error Alto)
0
2
4
6
8
10
Error (MSE)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
MSE de Reconstrucción
Umbral (P95)
Tiempo (s)

Notebook